Yu-Ting Liao (Harry)

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ABOUT

Designer and computational artist exploring the body as an interface through wearable technology, physical computing, and embodied sound and interaction. My practice combines fashion pattern cutting with computational methods, hardware prototyping, machine learning, and creative coding. I recently completed an MA in Computational Arts at Goldsmiths, building on a background in pattern cutting at UAL and generative-AI media at Shih Hsin University, with over three years of professional visual design experience in Taipei.

EDUCATION

Goldsmiths, University of London **London, United Kingdom**
MA in Computational Arts, Distinction **Sep 2025 – Sep 2026**

- **Final project:** Solo Choir.
- Physical computing, embodied interaction, real-time audio and machine learning for creative practice (Raspberry Pi Pico, Arduino, Python, p5.js, Tone.js, TouchDesigner).

Shih Hsin University **Taipei, Taiwan**
MA in Digital Multimedia Arts **Sep 2022 – Jul 2024**

- **Graduate thesis:** *Technology Inside AI Game Development and Application: An Example of Generative AI NPC*
  - Developed an AI NPC dialogue system in Unreal Engine 5.2 using GPT-4, integrating speech recognition, synthesis and sentiment analysis for context-aware, real-time NPC interaction.

University of the Arts London **London, United Kingdom**
BA in Fashion Pattern Cutting **Sep 2016 – Jun 2019**

SELECTED PROJECTS

Solo Choir **2026**
MA final project · real-time vocal instrument **Goldsmiths**

- Turns one singer into a choir. Sing a phrase and the system answers with harmony voices made by real-time neural voice conversion.
- Designed and built a tabletop responsive instrument (four speakers, head model, boom microphone) and two wearable prototypes, an open-ear bone-conduction headband and an auxetic neck ring for feeling harmony in the body.
- **Demo:** vimeo.com/1223687812 **Code:** github.com/yvwHY/solo-choir

Fabri-Form **2026**
Human–AI interface · pattern cutting **Goldsmiths**

- A translation tool between sketch and pattern. A multimodal LLM reads a hand-drawn sketch, fills a bounded parameter schema, and draws a printable 1:1 block in the browser. Six sliders expose every inferred parameter, so authorship stays with the designer. The current version outputs paper patterns.

Gesture-Float **2026**
Hardware · wireless control **Goldsmiths**

- A wireless flying unit steered by the arm. An arm-mounted Pico W streams hand pitch and roll over Wi-Fi (UDP) to a helium balloon unit, on hand-soldered circuits.

The Bonnet **2025**
Wearable · computer vision **Goldsmiths**

- An anti-surveillance headpiece. A custom-trained YOLOv8 model detects a CCTV camera and an Arduino servo drops a fabric veil over the face.

EXHIBITIONS

Scream, Unsynced · MA Computational Arts Festival 2026, Goldsmiths **London, Aug 2026**

- Postgraduate degree show, St James Hatcham Building, 27–29 Aug 2026. Exhibited Solo Choir.

Bye Default (Computational Arts Pop-Up), PSH Atrium, Goldsmiths **London, May 2026**

- Exhibited Harmonizing!, a browser-based live vocal-harmony instrument; designed the exhibition graphics and caption cards for all 37 works.

EXPERIENCE

Nice Crush Design House **Taipei, Taiwan**
Visual Designer **Sep 2023 – Jun 2025**

- Created visual assets, including digital advertisements, social media graphics, packaging and promotional materials.
- Developed marketing strategies to ensure cohesive visual communication across both online and offline platforms.

Digibionic **Taipei, Taiwan**
Visual Designer **Nov 2021 – Sep 2022**

- Designed and created visual assets for brand marketing.

AY-AY International **Taipei, Taiwan**
Graphic Designer **Dec 2020 – Jun 2021**

- Created graphic assets and layouts for a wide mix of multimedia projects.
- Produced and maintained style guides.

SKILLS

Making: pattern cutting, garment construction, circuit soldering
Physical Computing: Raspberry Pi Pico W, Arduino, servo/motor control, IMU orientation sensing, bone-conduction and surface-exciter audio
Audio / Machine Learning: real-time neural voice conversion, DDSP, pitch detection, Web Audio API, YOLOv8, OpenCV, Python, custom dataset curation (Roboflow)
Creative Code: p5.js, Tone.js, ml5.js, TouchDesigner, JavaScript
3D / Game Engine: Unreal Engine 5, 3D computer graphics
Design: Adobe Illustrator, Photoshop, InDesign, After Effects